

## 1. Administrative & Study Identification

**Full Study Title:** A Randomized, Double-blind, Placebo Controlled, 3-arm Multicenter Phase 3 Study to Assess the Efficacy and Safety of Ianalumab in Patients With Active Sjogren's Syndrome (NEPTUNUS-2) **Short Title/Acronym:** NEPTUNUS-2 **Protocol Number:** CVAY736A2302 (IRAS ID: 1005468) **NCT Number:** NCT05349214 **Sponsor Name:** Novartis Pharma AG **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase III **Medical Monitor:** Claus-Peter Danzer, Novartis Medical Monitor

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## 2. Study Synopsis & Rationale

**2.1 Study Objective** **Primary Objective:** To demonstrate the clinical efficacy, safety, and tolerability of ianalumab 300 mg administered subcutaneously monthly or every 3 months compared to placebo in patients with active Sjögren’s syndrome. **Secondary Objectives:** To evaluate the safety profile, B-cell depletion and recovery timelines, improvements in salivary flow, and reduction in systemic disease activity.

**2.2 Background & Rationale** Sjögren’s disease is a serious, progressive, systemic autoimmune disease with a significant detrimental impact on quality of life and very limited treatment options. Overactivation of B-cells plays a key role in the disease. Ianalumab employs a dual mechanism targeting B-cell depletion and B-cell Activating Factor (BAFF) receptor inhibition. This trial is conducted to provide an effective, targeted biological therapy to reduce disease activity for an established unmet medical need.

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## 3. Study Design & Methodology

**3.1 Design Framework** **Study Type:** Interventional **Control Type:** Placebo-controlled (3-arm study) **Blinding:** Double-blind (Quadruple-blinded structurally) **Randomization:** 1:1:1 (Ianalumab 300 mg monthly : Ianalumab 300 mg every 3 months : Matched Placebo)

**3.2 Planned Enrollment & Duration** **Target \$N\$:** ~504 patients **Number of Sites:** Multicenter (USA, UK, 10 EU countries, and other global sites) **Estimated Study Start:** 2022 **Estimated Primary Completion:** May-2025 (Study follow-up ongoing through Mar-2028)

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Women and men  $\geq 18$  years of age. 2. Classification of Sjögren’s syndrome according to the ACR/EULAR 2016 criteria with a diagnosis duration of  $\leq 7.5$  years at screening. 3. Positive anti-Ro/SSA antibody at screening and a screening ESSDAI score of  $\geq 5$  within 8 defined domains.

**4.2 Exclusion Criteria** 1. Pregnant or nursing (lactating) women, or women of child-bearing potential not using highly effective contraception. 2. Evidence of active tuberculosis (TB) infection. 3. Positive for Hepatitis B (HBsAg) or Hepatitis C (without spontaneous clearance).

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Ianalumab 300 mg administered once monthly OR once every 3 months (vs. matched placebo). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 52 weeks (1 year) of active treatment, followed by a post-treatment safety follow-up period of up to 2 years.

**5.2 Concomitant & Prohibited Therapy Permitted:** Stable doses of usual eye drops, artificial tears/saliva, salivary stimulants, hydroxychloroquine ( $\leq 400$  mg/day), methotrexate ( $\leq 25$  mg/week), azathioprine ( $\leq 150$  mg/day), and systemic corticosteroids ( $\leq 10$  mg/day prednisone equivalent). **Prohibited:** Other investigational biologicals, unapproved targeted immunosuppressive therapies, or initiation of new targeted disease-modifying therapies during the trial.

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days)   | Key Activities/Procedures                                                                   |
|----------|-----------------|---------------------------------------------------------------------------------------------|
| 1        | Day -28 to 0    | Informed Consent, Medical History, ESSDAI Scoring, Anti-Ro/SSA Labs                         |
| 2        | Day 0           | Randomization, First SC Dose, Safety Audit                                                  |
| 3        | Weeks 4 to 52   | Monthly visits (~13 times) for IP administration, Efficacy Assessment, Safety Labs          |
| 4        | Weeks 52 to 156 | Post-treatment monitoring for B-cell recovery (monthly for 5 visits, then every 3-6 months) |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Change from baseline in the EULAR Sjögren's syndrome disease activity index (ESSDAI) score at Week 52. **Measurement Method:** Clinical assessment across 8 physiological domains (constitutional, lymphadenopathy,

glandular, articular, cutaneous, renal, hematological, and biologic).

**7.2 Secondary & Exploratory Endpoints** \* Safety and tolerability profile (Frequency of AEs/SAEs over 3 years). \* Change in stimulated/unstimulated salivary flow. \* Pharmacodynamics: B-cell depletion rates and post-treatment B-cell recovery timelines.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Rigorous monthly monitoring during the treatment phase; specific observation for infections and injection-site reactions. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** External committee overseeing trial safety, especially regarding immunosuppression risks.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy; Safety Set for all patients receiving at least one dose of IP. **Sample Size Justification:** Powered for a 3-arm superiority comparison of ionalumab regimens versus placebo ( $\alpha = 0.05$ ). **Handling of Missing Data:** Multiple Imputation (MI) and model-based approaches for longitudinal data drops during the 3-year span.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite monitoring visits coupled with centralized data review. **Audit Trail:** Continuous eCRF locking after primary completion and systematic data clarification forms.

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## 11. Study Identifiers & Governance

**Primary ID:** CVAY736A2302 **Registry Number:** NCT05349214 **Sponsor/Lead Organization:** Novartis Pharma AG **Study Title:** NEPTUNUS-2 **Official Regulatory Title:** A Randomized, Double-blind, Placebo Controlled, 3-arm Multicenter Phase 3 Study to Assess the Efficacy and Safety of Ianalumab in Patients With Active Sjogren's Syndrome

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## 12. Clinical Framework (Sjögren's Specific)

**Primary Condition:** Active Sjögren's Syndrome **Diagnosis Threshold:** ACR/EULAR 2016 criteria & ESSDAI score  $\geq 5$  **Medical Methodology:** Monoclonal antibody therapy (B-cell depletion & BAFF receptor inhibition) **Optimization Target:** Reduction in systemic disease activity (ESSDAI score improvement)

### 13. Study Procedures & Methodology

**Management Pathway:** Standard of care + experimental targeted biological therapy **Education Protocol (HCP):** Subcutaneous injection administration and B-cell monitoring protocols **Education Protocol (Patient):** Symptom tracking and infection-risk awareness **Quality Audit Mechanism:** HCP ESSDAI scoring standardization & central lab verification

## 14. Observation & Follow-Up Schedule

**Total Duration:** 3 Years (52 weeks treatment + 104 weeks follow-up) **Onsite Visit Cadence:** Monthly for Year 1, monthly for first 5 months of follow-up, then every 3-6 months. **Remote Monitoring Frequency:** Intermittent safety follow-up calls as required **Checklist Review Interval:** Every study visit (Q4W during treatment)

## 15. Sign-off & Approval

| Role                     | Name          | Signature | Date          | --- | --- | --- | --- |
|--------------------------|---------------|-----------|---------------|-----|-----|-----|-----|
| Principal Investigator   | [Insert Name] | _____     | [DD-MMM-YYYY] |     |     |     |     |
| Clinical Operations Lead | [Insert Name] | _____     | [DD-MMM-YYYY] |     |     |     |     |
| Lead Statistician        | [Insert Name] | _____     | [DD-MMM-YYYY] |     |     |     |     |